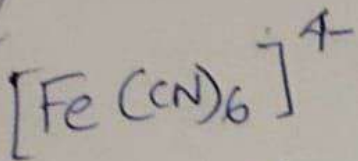
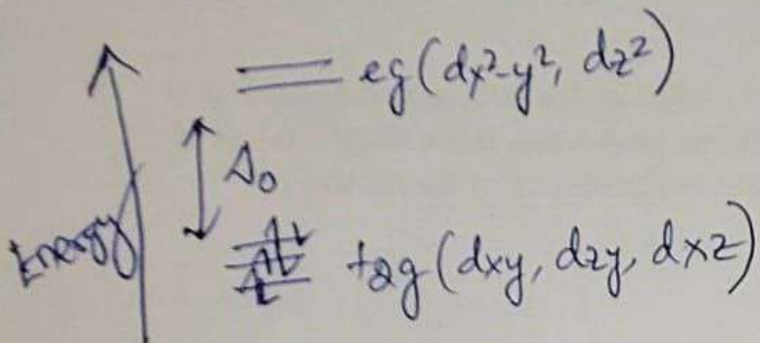


①



$\text{Fe}^{2+} \rightarrow d^6$ $\text{CN}^- \rightarrow$ strong field ligand $\rightarrow$ Low spin.
($t_{2g}^6 e_g^0$)



$\text{CFSE} = 6(-0.4\Delta_o) = -2.4\Delta_o$

$E = \frac{hc}{\lambda} = 6.513 \times 10^{-19} \text{ J}$

$N_A = 6.023 \times 10^{23}$

$E / \text{molecule} = N_A \times E$

$= 392 \text{ KJ/mol}$ (CFSE = -9408 KJ/mol)

$N_A = 6.314 \times 10^{23} \rightarrow E = 411 \text{ KJ/mol}$

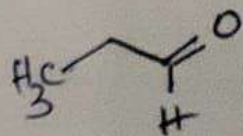
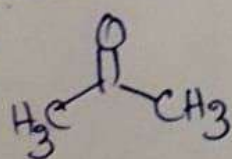
$\text{CFSE} = -986 \text{ KJ/mol}$

② $\text{HI} \rightarrow$ strong acid $[\text{H}^+] = 0.03091$

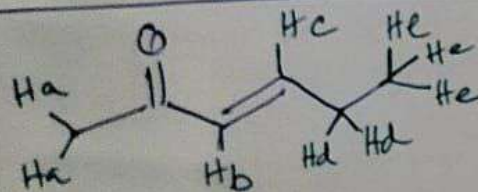
$\text{pH} = -\log [\text{H}^+] = -\log(0.03091) = \underline{\underline{1.51}} \sim \underline{\underline{1.5}}$

$\text{pOH} = 14 - \text{pH} = \underline{\underline{12.49}} \sim \underline{\underline{12.5}}$

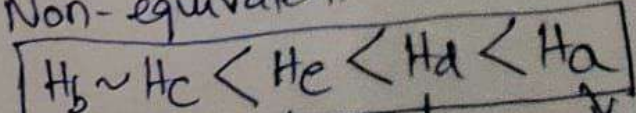
③



Aldol
major pdt

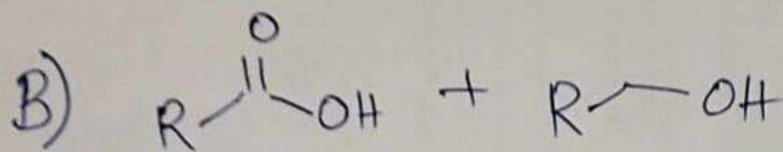
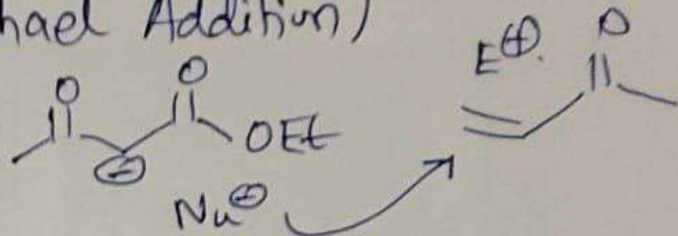
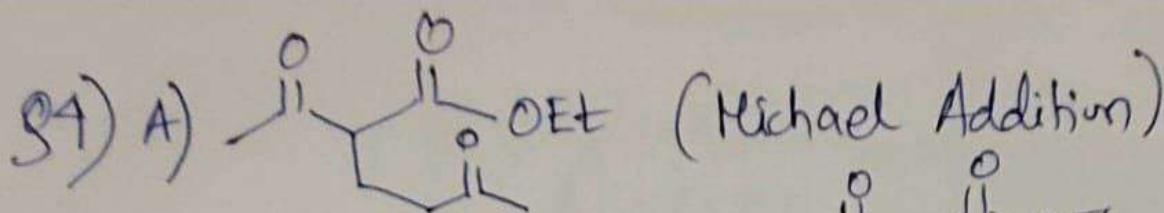


Non-equivalent H $\rightarrow$ Ha/Hb/Hc/Hd/He

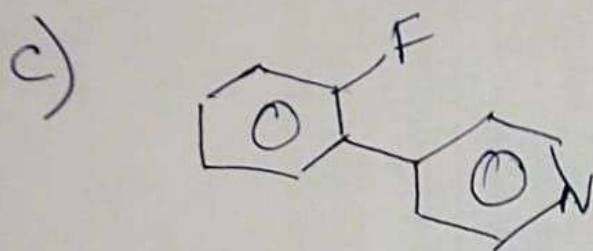


Increasing order

vinyllic alkyl allylic α -carbonyl

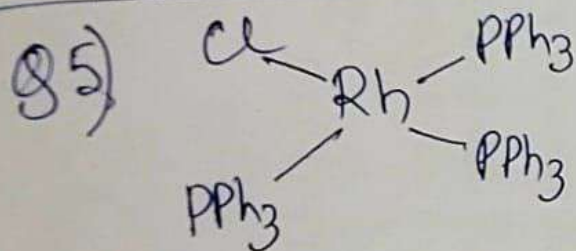


Cannizzaro $\rightarrow$ acid + alcohol.

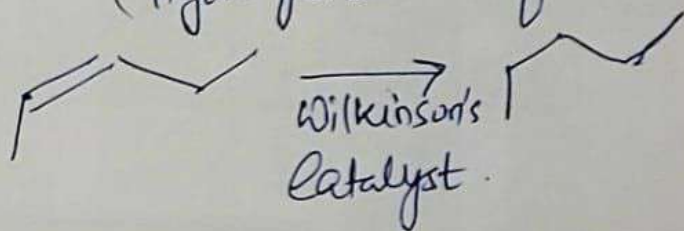


(Suzuki coupling)

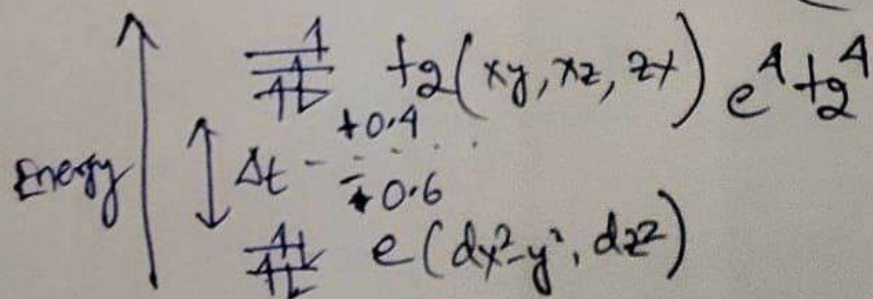
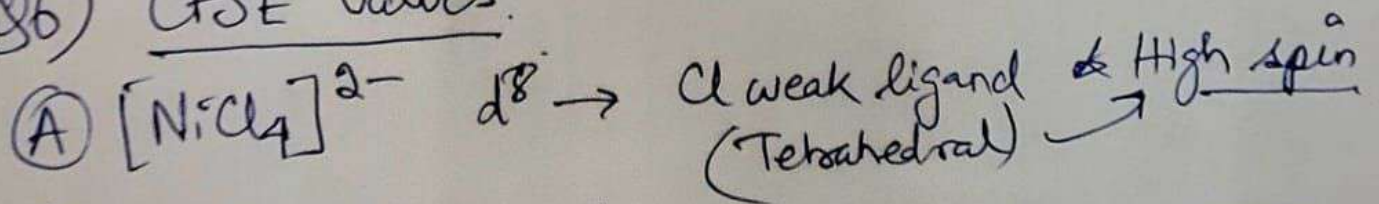
Scored half if minor products have been shown



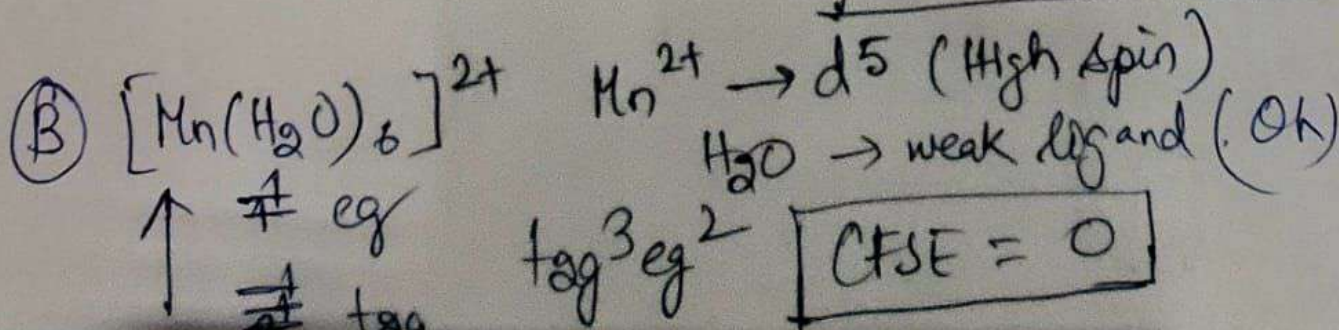
(Hydrogenation of alkenes)

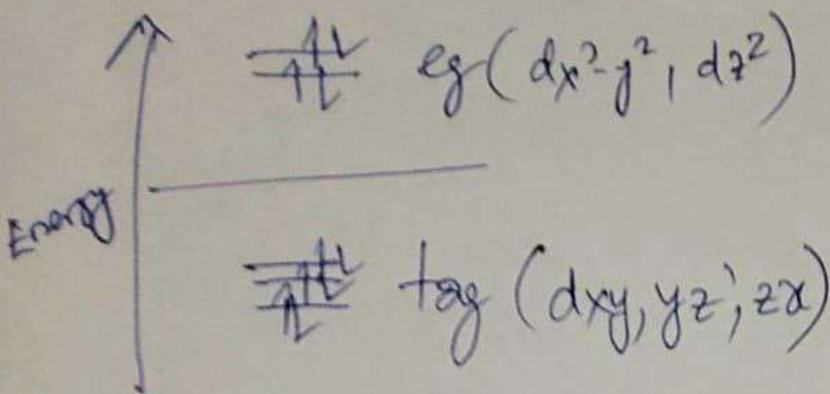
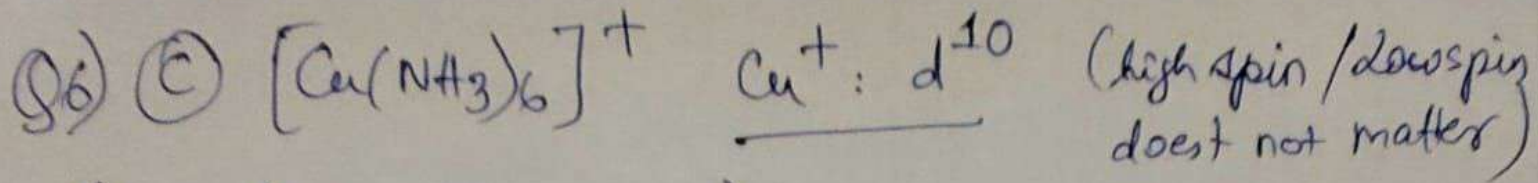


86) CFSE values.



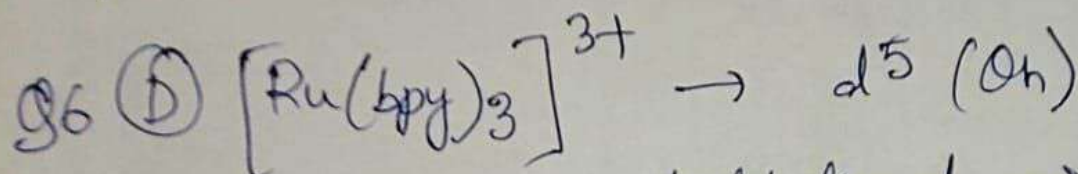
$$\begin{aligned} \text{CFSE} &= -0.8 \Delta_t \\ &= (-0.8 \times \frac{4}{9} \Delta_0) \\ &= -0.35 \Delta_0 \end{aligned}$$



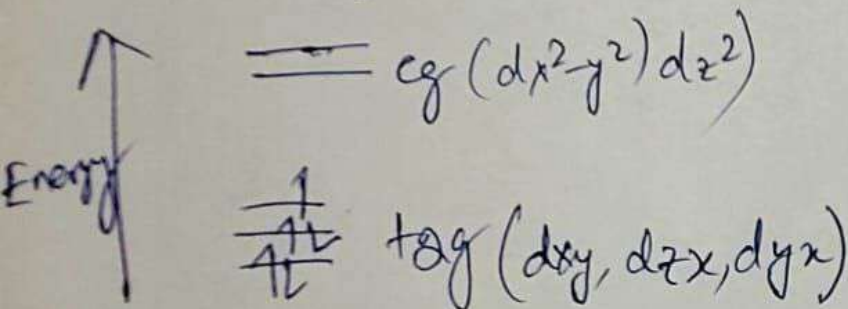


$\boxed{\text{CFSE} = 0}$

$\boxed{t_{2g}^6 e_g^4}$



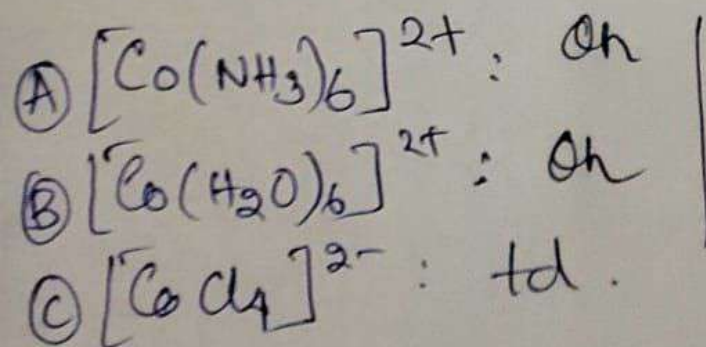
bpy $\rightarrow$ strong field ligand $\rightarrow$ low spin.



$\boxed{\text{CFSE} = -0.4 \times 5 = -2.0 \Delta_0}$

Q7) $E = hc/\lambda$ therefore $E \propto 1/\lambda$ ($\lambda \propto 1/E$)

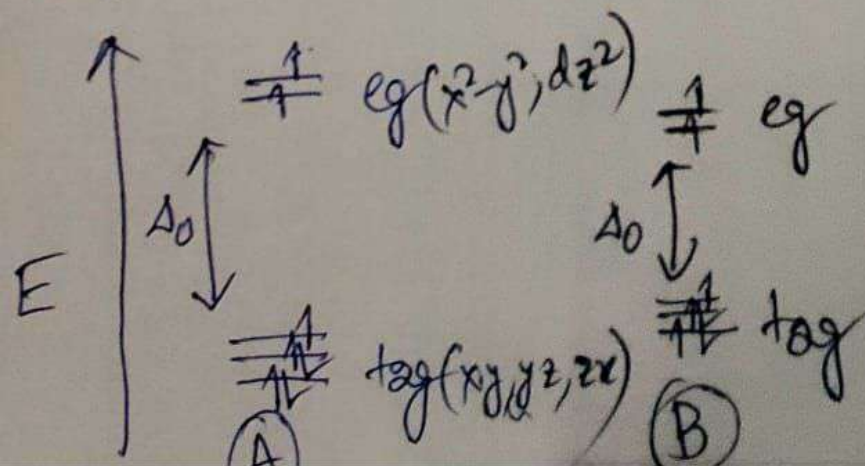
Inverse Relation.



d^7 system (HS) complexes.

$\boxed{\Delta_0(\text{A}) > \Delta_0(\text{B}) > \Delta_t(\text{C})}$

$\boxed{\lambda_A < \lambda_B < \lambda_C}$



$\Delta_t \approx (4/9 \Delta_0)$

(C)

Complex	Geometry	Ligand Strength	Δ	λ_{abs}	$\lambda_{emitted}$
$[\text{Co}(\text{NH}_3)_6]^{2+}$	Oh	Strong	Large	Blue/Violet	Yellow
$[\text{Co}(\text{H}_2\text{O})_6]^{2+}$	Oh	Medium	Moderate	Green	Pink
$[\text{CoCl}_4]^{2-}$	Td	Weak	Small	Red/Orange	Blue

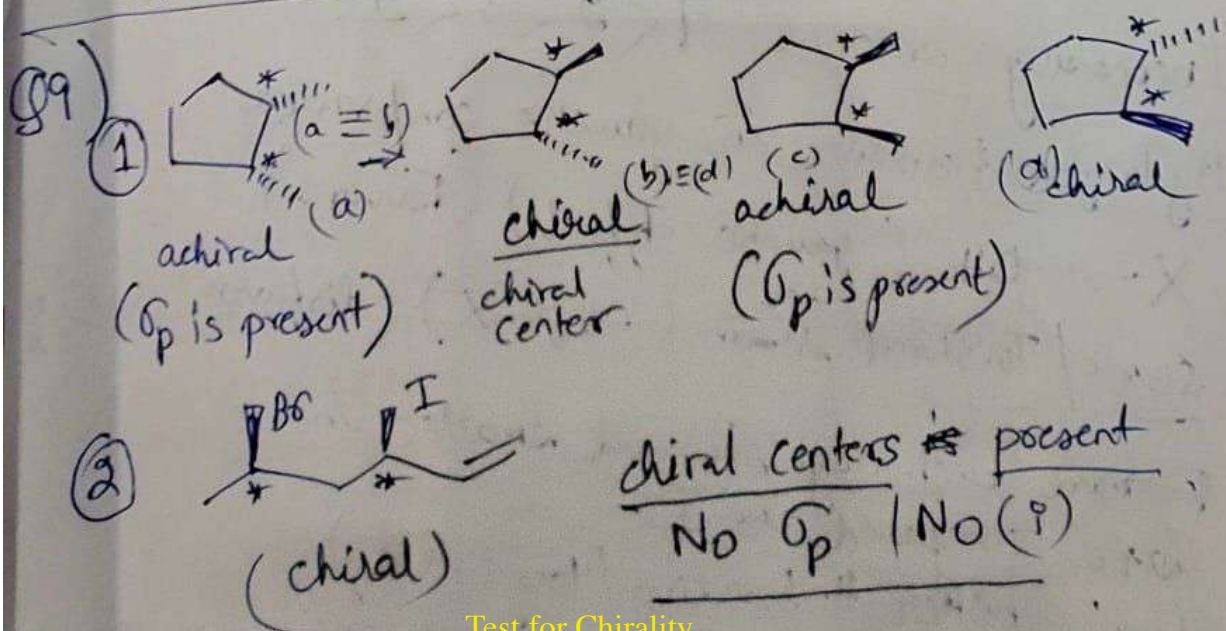
$\lambda_{emitted}$ is complementary color - VIBGYOR.
 $\rightarrow$ color of complexes.

Q8) Refer Notes $\rightarrow$

PDT $\rightarrow$ generates ROS from excited triplet state
 ROS kills cancer cells.

Photocaging $\rightarrow$ light releases chemoreactive drug by photoinduced bond-breaking $\rightarrow$ kill cancer cells.

Light controls activation and killing.

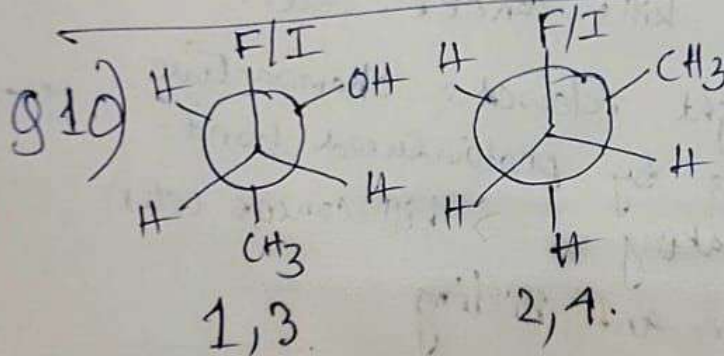
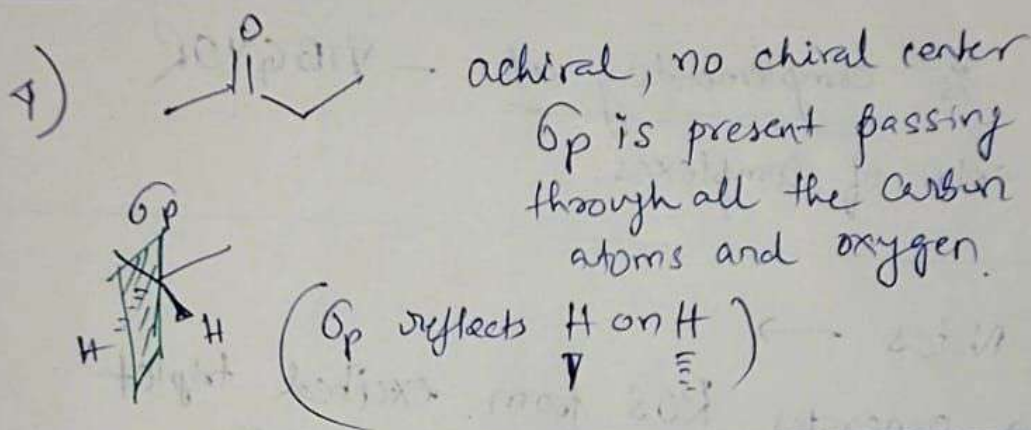
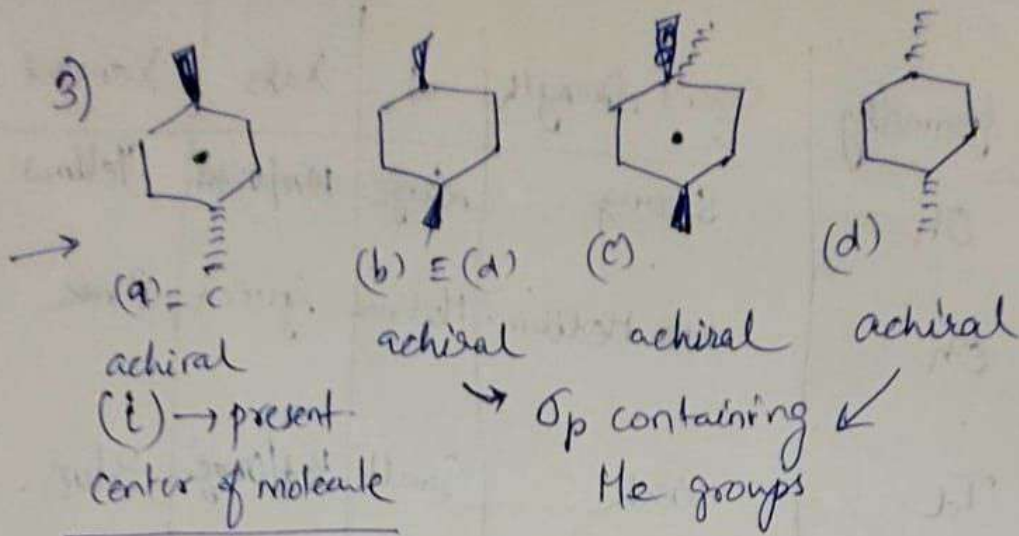


Test for Chirality

Non-superimposable mirror images

No POS/COS

These points have to be mentioned for scoring full marks



Increasing order : $1 < 2 < 3 < I$.

⊕ Hydrogen Bonding in gauche conformer

$X \cdots OH$ (gauche effect) (Favors)

⊕ Steric/Torsional strain (CH_3-X) (disfavors)

F more electronegative and smaller in size compared to I

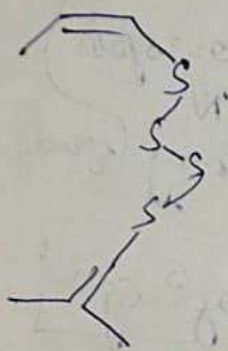
F favors better H-bonding/less steric

I $\rightarrow$ creates more steric strain/

for lower H-bond capacity

Q11)
 Fe: Hb / Mb / cytochromes / catalase etc.
 Cu: SOD / cytochromes / amino oxidases /
 Zn: carbonic anhydrase / carboxy peptidase /
 Ni: Nickel hydrogenase / Urease, Alcohol dehydrogenase
 Ni-Fe-Se hydrogenase

Q12) Vulcanization: heating crude rubber in S_2Cl_2 in CS_2

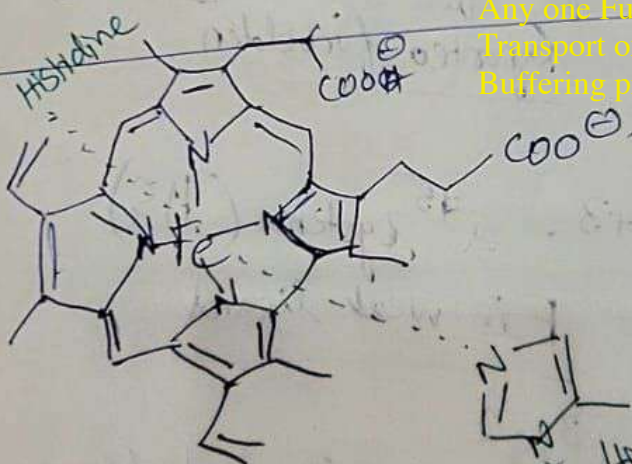


S-S disulfide crosslinkage

Increases elasticity and ability to regain shape
 Increases strength

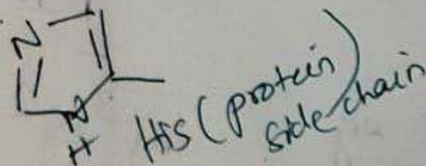
Any one Function of Hemoglobin (0.5 Marks):
 Transport oxygen from lungs to tissues;
 Buffering property

Q13)



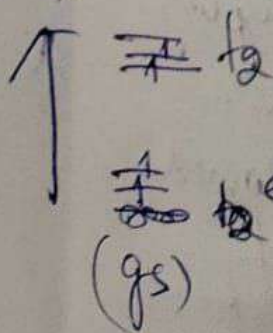
proximal & distal histidines

Refer Notes.



(Sickle cell anemia / porphyria / anemia)

Q14) $[MnCl_4]^- \rightarrow Td (Mn^{+3}: d^4)$
 (Always high spin) $(e^2 t_2^2)$



Spin ~~forbidden~~ allowed.
 Laporte allowed

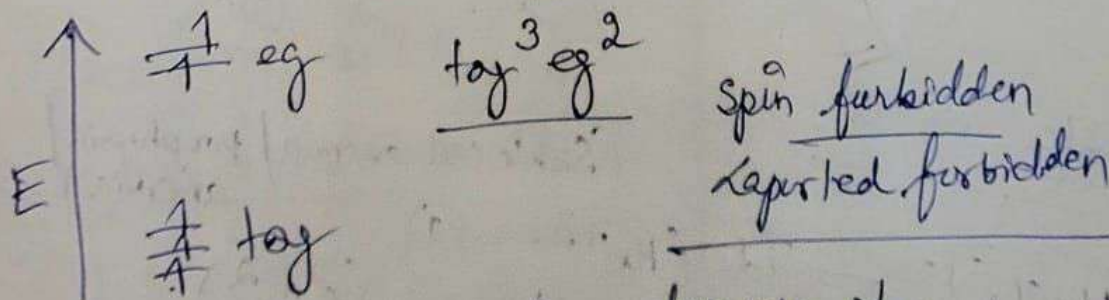
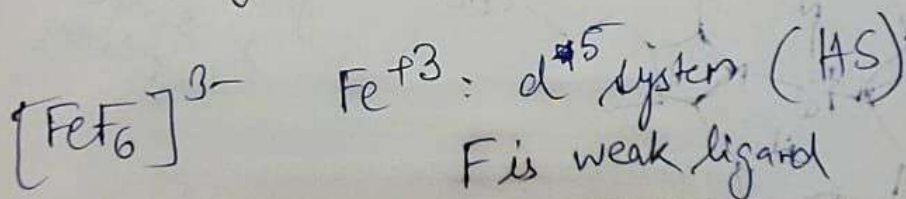
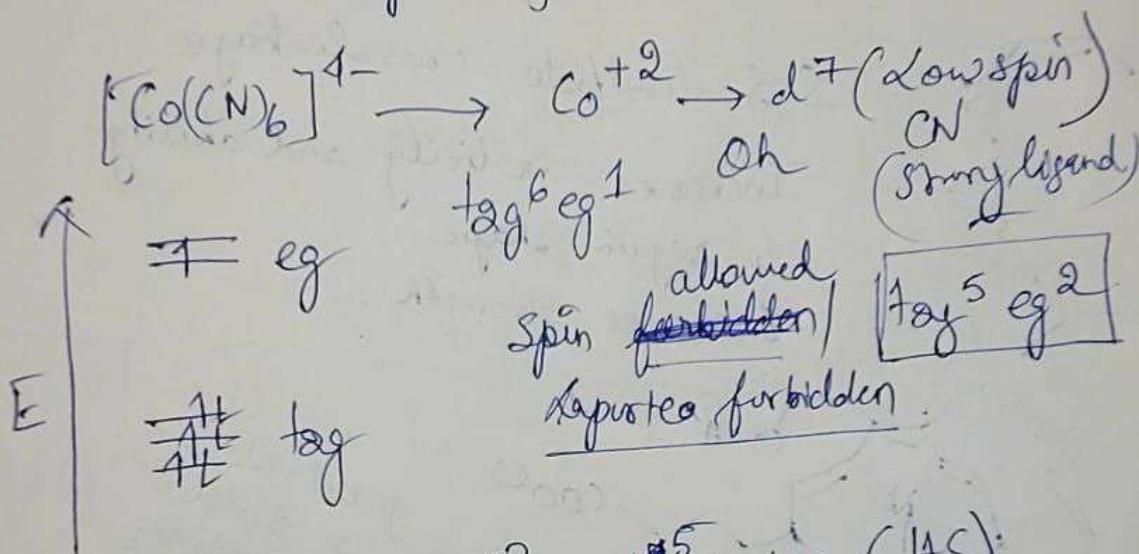
$\rightarrow e^1 t_2^3$
 (excited state)

Extraction coefficient $\propto$ probability of transition $\rightarrow$ how to what extent the transition is allowed.

Transitions (d-d) are governed by selection rules (spin selection $\Delta S = 0$) allowed.

Laporte selection rule: $g \rightarrow g$ $\times$
(centrosymmetric molecule) $u \rightarrow u$

Oh $\rightarrow$ Laporte forbidden



Some Relaxation occurs because of spin-orbit coupling & vibronic coupling. So we see transitions (but weak).

$$\epsilon_{\text{Fe}} < \epsilon_{\text{Co}} < \epsilon_{\text{Mn}}$$

More allowed transitions / higher is ϵ values.

$$g15) \quad +125^\circ \rightarrow R$$

$$-125^\circ \rightarrow S$$

$$\text{composition } x = R \quad (1-x) = S.$$

$$x \times 125 + (1-x) 125 = 100.$$

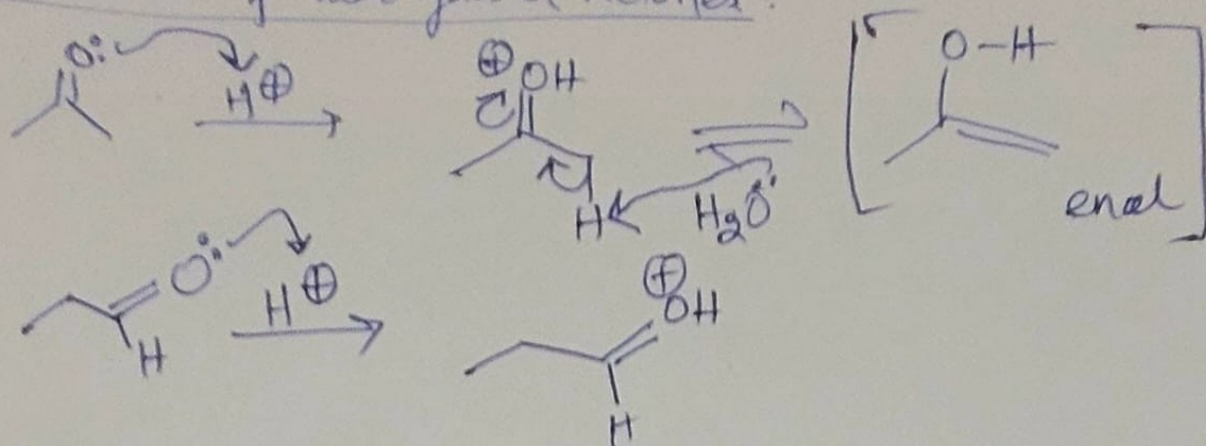
$$x = 90\% (R) \quad (S) = 10\%$$

$$ee = (90 - 10) = 80\%.$$

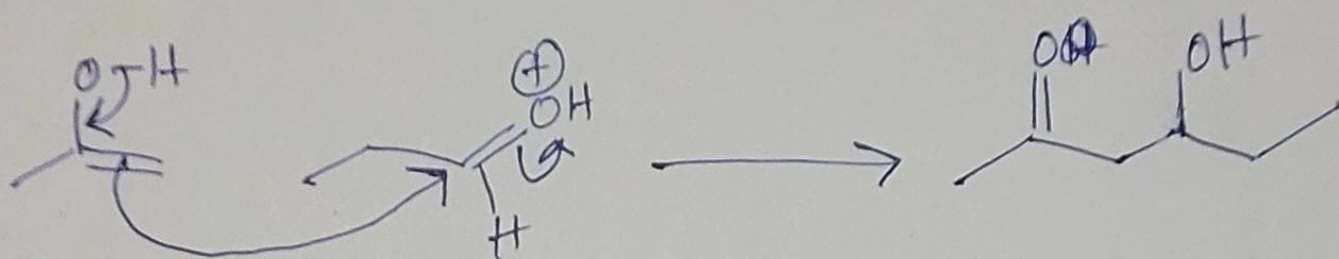
$$(\text{Other way}) \Rightarrow \boxed{ee = \frac{100}{125} \times 100 = 80\%}$$

$\left(\frac{\text{observed Rotation}}{\text{Absolute Rotation}} \right) \times 100$

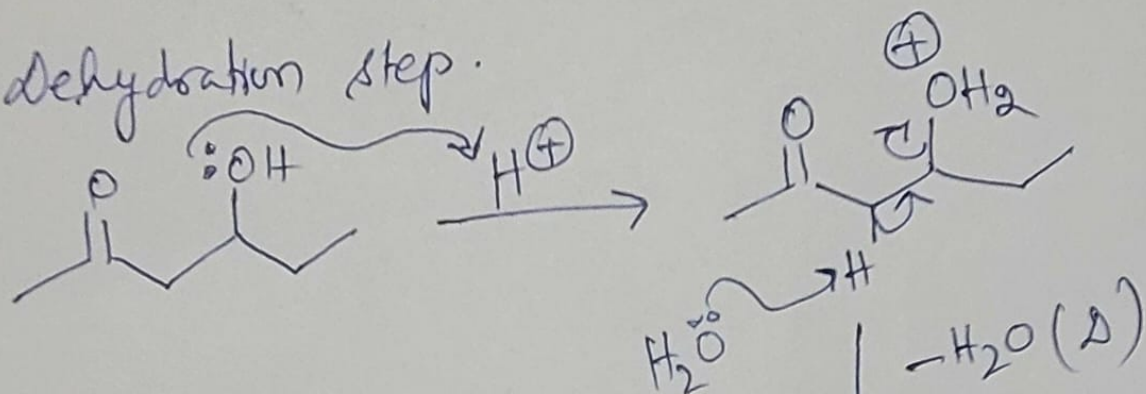
1) Protonation of aldehydes & ketones:



2) Nucleophilic Attack on protonated propanal E^+ Carbonyl center.



3) dehydration step.



Final product

